Exp No: 10 a) Internetworking with Routers in
Date: 01/10/26 CISCO Packet Tracer Simulator.

Design and configure a simple Internetwork using a Router.

AIM:

To implement and design configure a simple internetwork using a router and two PCs in CISCO packet Tracer and Verify network connectivity.

Procedure:

* Open Cisco packet Tracer and create a new network topology.
* place one router and 2 PCs in the workspace.
* Connect PC0 and PC1 to the router using copper Straight - Through Cables.
* configure Router 1 Interfaces with the following
  * FastEthernet 0/0 → 192.168.10.1/24
  * Fast Ethernet 0/1 → 192.168.20.1/24
* Enable both router interfaces using the no shutdown command.
* configure PC0 with
  - IP Address → 192.168.10.2
  - Subnet Mask → 255.255.255.0
  - Default gateway → 192.168.10.1
* Configure PC1 with:
  - IP Address → 192.168.20.2
  - Subnet Mask → 255.255.255.0
  - Default gateway → 192.168.20.1

* Open the Simulation Mode in Packet Tracer.

* Send a simple PDU from PC0 to PC1

* Verify that the PDU from each PC successfully and an acknowledgment is Received



Result:

A simple Internetwork using a Router and two PCs was successfully designed and configure.

Exp. No: 10
Date: 01/10/26

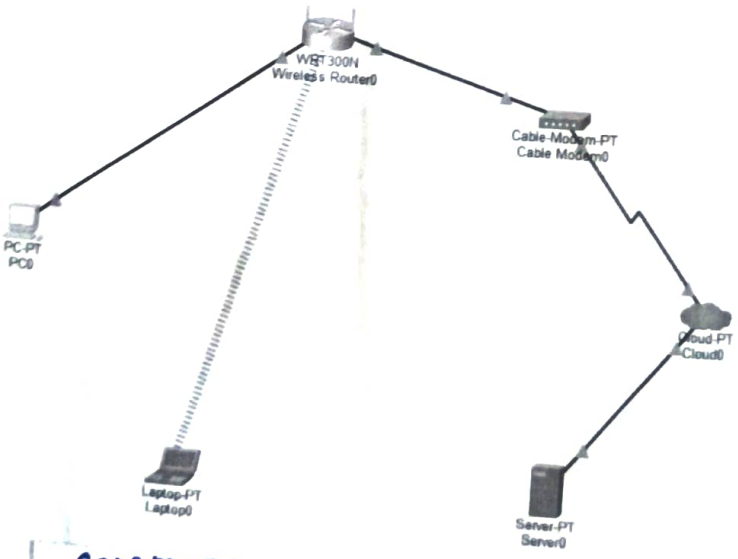
b) Design and configure an internetwork using wireless router, DHCP & Internet cloud.

AIM:

To design & configure an internetwork using a wireless Router, DHCP Server, & Internet cloud and verify Network connectivity.

Procedure:

* Open cisco packet Tracer and create a new network topology.
* Add a wireless Router, PC, laptop, cable modem, Internet cloud, and cisco.com Server.
* Connect the device using Appropriate copper straight through and coaxial cables.
* configure the device using Appropriate cable. IP address 192.168.0.1 and enable the DHCP Server.
* Set the Wireless Network name (SSID) as



with a wireless its IP address

is Services on

using the

command ping

Result:

A Internetworking using wireless Router, DHCP Server & Internet cloud was Successfully designed

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Design and configure an Internetwork using wireless router, DHCP & Internet cloud.

AIM:

To design & configure an Internetwork using a wireless Router, DHCP Server, & Internet cloud and verify Network connectivity.

Procedure:

* Open cisco packet Tracer and create a new network topology.
* Add a wireless Router, PC, laptop, cable modem, Internet cloud, and cisco.com Server.
* Connect the device using Appropriate copper straight through and coaxial cables.
* Configure the device using Appropriate cable. IP address 192.168.0.1 and enable the DHCP Server.
* Set the Wireless Network name (SSID) as Home Network.
* Configure the laptop with a wireless module and the PC to obtain its IP address automatically using DHCP
* Enable DHCP and DNS Services on the cisco.com Server
* Test connectivity by using the command ping cisco.com.

Result:

A Internetworking using wireless Router, DHCP Server & Internet cloud was Successfully designed